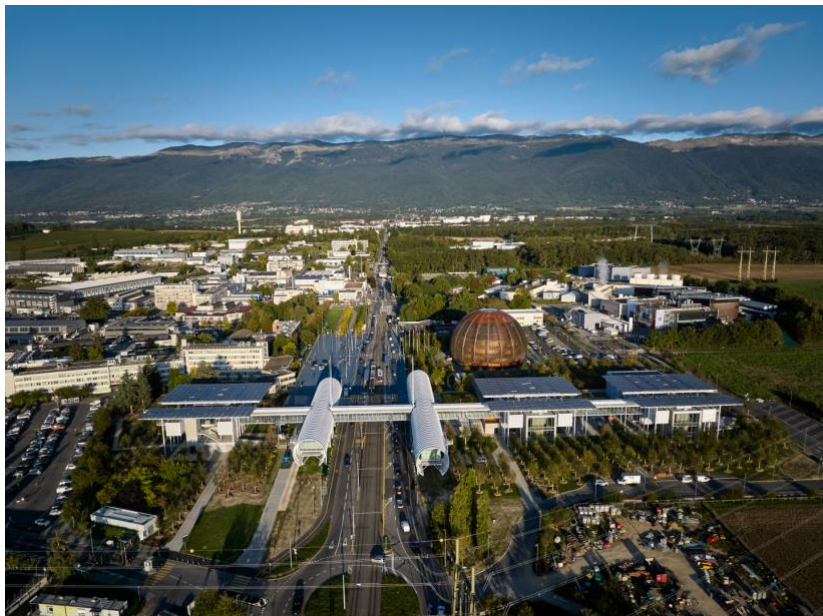


Managing energy responsibly: CERN passes ISO 50001 audit

CERN is one of the first scientific laboratories to have obtained ISO 50001 certification for energy management. Following our first surveillance audit, find out how we can all play a role in improving energy management



The CERN Meyrin site. Continual improvement of energy management is one of the key pillars of the Organization's strategy to minimise its impact on the environment. (Image: CERN)

All members of the CERN community are ambassadors for CERN's commitment to environmentally responsible research. We therefore need to understand the implications and audits of the ISO 50001 certification – the international standard used to continually monitor and improve energy management.

CERN was awarded the ISO 50001 certification on 2 February 2023 for a period of three years, covering the entire perimeter of the Organization: all sites, activities and energies. CERN's unique array of accelerators, detectors and infrastructure is primarily powered by electricity, accounting for about 95% of CERN's total energy use. In addition, the Laboratory uses natural gas for heating, fuel for its fleet of vehicles, diesel for emergency generators and commercial liquid nitrogen for cooling. The ISO 50001 standard provides guidance and tools to improve energy performance and integrate energy management into our overall efforts to improve quality and environmental management.

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The May/June issue of the CERN Courier is out

In this context, mandatory annual surveillance audits are carried out by the French Association for Standardization, AFNOR. The first surveillance audit for CERN took place from 28 February to 1 March. It required weeks of preparation, including an internal pre-audit with EDF in January.

The process included technical assessments of the Laboratory's largest energy consumers. Focused interviews were held with Management, technical teams including cryogenics, electricity, cooling and ventilation, and support teams such as HR, procurement and HSE. Training and awareness raising of the entire CERN community is an important facet of the certification; a dedicated communications plan was therefore also assessed during the audit.

Conclusions were positive and identified no non-conformities. The auditor produced an official report at the end of April recommending that CERN maintain the certification without reserve: "Although recently deployed, the energy management system demonstrates a good level of maturity and is based on a well-structured

organisation. Changes have been implemented since last year, both in the provision of resources and in the organisation of new steering bodies in conjunction with the energy team."

"The conclusion of the audit highlights how well the requirements of the standard have been implemented across the Organization, from top management to technical teams," notes CERN's energy coordinator, Nicolas Bellegarde. "The auditor was particularly impressed with the energy improvement actions carried out by CERN's services and official bodies in 2023, including the experiments, demonstrating that the overall energy performance has improved and remains a top priority for the Organization." The next audit will be carried out in February 2025.

Share your feedback about energy saving and find out more about CERN's approach to energy management and the ISO 50001 certification here: <https://hse.cern/content/energy-management>.

HSE unit

Accelerator Report: Already a fifth of the way there

The whole CERN accelerator complex and its associated experimental facilities have been fully operational for some time now, so the time is ripe to review the first part of this year's run and look forward to what is still to come.

At the LHC, first collisions with just a few bunches occurred on 6 April. Meaningful physics data taking can only begin when collisions take place with at least 1200 bunches per beam, and this milestone was reached on 14 April. It means that, out of the 147 days allocated to proton-proton collisions this year, 32 have already been completed, representing just over 20% of the 2024 proton run.

During these initial 32 days, the LHC machine was available 67.2% of the time with stable beams in collision 45.2% of the time. The goal is to achieve a stable beam time ratio of at least 50%. Such figures are quite normal in the early stages of an annual run, a period during which various teething problems and challenges are identified and addressed, as discussed [in my last report](#).

Luminosity production is also progressing according to schedule, as can be seen in the graph below. So far, the integrated luminosity collected has reached 17.5 fb^{-1} , which is nearly 20% of the 2024 target of 90 fb^{-1} . To reach this target, we need an average of about 0.8 fb^{-1} per day. Recently we have seen a record production of 1.23 fb^{-1} in just 24 hours, which demonstrates the LHC's impressive potential to meet and possibly even exceed our target.



The predicted and achieved luminosity curves for ATLAS and CMS. The blue areas indicate the machine development (MD) periods, the red area the Van de Meer run, and the green one the technical stop. (Image: CERN)

Today, 15 May, the LHC is performing the Van de Meer run, [a crucial process](#) used to calibrate the luminosity measurements in the four main LHC experiments (ALICE, ATLAS, CMS, LHCb). This follows the first block of machine development (MD) sessions that took place on 13-15 May, during which experts conducted various tests and studies, including further investigation into [the collimator hierarchy breaking](#) that occurred in April.

On 18 May, the LHC will resume its primary task of producing luminosity. This production period will continue until the second MD block that is scheduled to start on 5 June and will be followed by a 5-day technical stop to carry out essential preventive and corrective maintenance before the summer holiday season. During the summer, the LHC will continue its luminosity production.

Not only is the LHC performing well, but the injector complex is also achieving a high level of beam availability for its experimental users. Physics in the injector complex kicked off on 22 March in the PS East Area. With the run scheduled to end on 2 December, the East Area has already completed over 20% of its 2024 run. Meanwhile the antimatter factory, which was the last facility in the injector complex to start beam operation, began delivering antiprotons to its experiments on 22 April, so just over 10% of its scheduled physics time for 2024 has now elapsed.

Linac4, the first link in the proton injector chain, has posted an average availability of 97.3% since its start after Long Shutdown 2 (LS2). It has posted

availability of 95.7% so far in 2024, about 1.6% less than usual. This can mainly be attributed to a series of faults that led to [the replacement of a klystron](#) in March.

Facility	Destination	2024 Overall [%]	2024 by destination [%]	Period
LINAC4	PSB	95.7	95.7	15.02.2024 – 14.05.2024
PSB	PS	94.2	94.2	21.02.2024 – 14.05.2024
	ISOLDE		96.0	28.03.2024 – 14.05.2024
PS	SPS	90.6	91.3	01.03.2024 – 14.05.2024
	East Area		91.7	22.03.2024 – 14.05.2024
	nTOF		92.2	25.03.2024 – 14.05.2024
	AD		92.2	14.03.2024 – 14.05.2024
SPS	LHC	85.9%	97.2	08.04.2024 – 14.05.2024
	North Area		86.2	25.03.2024 – 14.05.2024
	AWAKE		97.2	15.04.2024 – 28.04.2024
	HiRadMat		93.2	29.04.2024 – 04.05.2024

An overview of the beam availability of the different injectors and experimental facilities. The availability of each machine takes into account the non-availability of the upstream machines. (Image: CERN)

The injectors will continue their routine beam delivery to the experimental facilities until 12 June, when they too will interrupt beam production for a 4-day technical stop. Afterwards, it will be business as usual once more and the experiments can look forward to good beam delivery over the summer.

As we continue through the year, the achievements so far set a promising pace for the remaining months.

Rende Steerenberg

CERN70: An electronic revolution

Part 9 of the CERN70 series. Georges Charpak received the 1992 Nobel Prize in Physics for his 1968 invention of the multi-wire proportional chamber, which revolutionised particle detection

Progress in fundamental particle physics research often relies on developing ever more precise detectors with increased reaction times and covering increasingly larger surfaces. Early detectors at CERN fulfilled only part of these requirements. They were mainly based on photographic methods unsuitable for studying rare particle interactions, where only a few interesting events must be selected out of millions observed. It was necessary to increase the speed of data collection dramatically.

In 1968, Georges Charpak used an earlier development, the proportional counter, to build a new kind of detector, the “multiwire proportional chamber”. In a proportional counter, an electrical voltage is applied to the terminals of a gas-filled tube with a central wire running through it. As particles pass through, the gas is ionised, triggering a chain reaction that generates an electrical signal on the wire. This signal is used to determine the location of the first ionisation.



From left to right, Georges Charpak, Fabio Sauli and Jean-Claude Santiard working on a “multiwire proportional chamber” in 1970. This chamber, developed by Charpak in 1968, was a gas-filled box with many parallel detector wires, each connected to individual amplifiers. Linked to a computer, it could achieve a counting rate a thousand times better than existing techniques – without a camera in sight. (Image: CERN)

Instead of a tube and a single wire, Charpak’s idea was to use a gas-filled enclosure containing many parallel detection wires. Each wire connects to a transistor amplifier and acts as an independent proportional counter. This device, linked to a computer, achieved a counting speed a thousand times faster than the detectors of the time and a spatial resolution of the order of a millimetre. A later development, the drift chamber, achieved an even better spatial resolution of fractions of a millimetre. Charpak’s invention revolutionised particle detection, bringing it into the electronic age.

Today practically every experiment in particle physics uses some type of track detector based on the principles of multiwire proportional chambers. These kinds of detectors are also used in many other fields, such as biology, radiology and nuclear medicine.

Recollections

Georges Charpak joined CERN in 1959. During his career, he worked on various experiments and developed numerous detectors that had a fundamental impact on particle physics. In 1992, he was awarded the Nobel Prize in Physics for his invention and development of particle detectors, in particular the multiwire proportional chamber.

“I almost never started to study a detector without having to solve a problem for my own experiments or, very often, to overcome difficulties confronting the community in which I was working. One of these problems, in the 1960s, was that the master tools in particle physics, the bubble chambers, were reaching a bottleneck. With about 10 million pictures taken every year, the high-energy physics community around the world reached the

maximum possible amount of photographs that could be reasonably treated by human examination.



Georges Charpak (seen here in 1978) won the 1992 Nobel Prize in Physics “for his invention and development of particle detectors, in particular the multiwire proportional chamber”. (Image: CERN)

Other detectors, the spark chambers, could be triggered by fast detectors to select interactions imbedded in a large flow of uninteresting ones, but this technique reached the same limit of millions of pictures to analyse. This is when I left a successfully completed experiment, the measurement of the muon magnetic moment, for which I had come to CERN, and thought of a method to retrieve the position of a spark in a spark chamber without having to take a picture [...].

The revolution in transistor amplifiers permitted the conception of a new idea [...]. With the experience I had gained earlier in the construction of single-wire proportional counters for my thesis in low energy nuclear physics, I conceived a small chamber of 10 by 10 centimetres with simple ingredients that made each wire operate in the same conditions as in a single wire proportional counter. When I tested the chamber, I observed pulses proportional to the energy lost in the gas by traversing particles, with several properties

opening the door to innovative detectors: the pulses on the anode wires were not due to the collection of the electrons of the avalanches, but to the motion of the positive ions inducing pulses of opposite polarity on the neighbouring wires. This made it easy to identify the wire collecting the avalanche. A counting rate of one million pulses per second, 1000 times larger than the one reached with spark chambers could easily be obtained.

The pulses were delayed with respect to the initial particles by a time of a few nanoseconds, which easily allowed measurement of the distance of the trajectory to the wire. This opened the way to another major advance: the drift chambers, which permitted the construction of large detecting surfaces with a small number of electronic channels and accuracies of the order of 100 microns. The positive induced pulses on the cathodes permitted the construction of two-dimensional detectors, essential for the localization of X-rays or neutrons.

I started to be also interested in the application of detectors to fields outside particle physics. In 1974, my group, in collaboration with Saclay, undertook the exploration of the human body with high energy particle beams. With wire chambers giving the position of incoming and outgoing particles, we had the idea of mapping the three-dimensional density distribution inside a living body without having to rotate either the source, the body or the detector. After good first results, we halted the project since no low-cost accelerators for this application were in view.

Nevertheless, the existence of groups active in detector research in high-energy physics can be a source of important achievements for other fields – if experienced people have the freedom also to tackle problems not directly connected with the main goal of their laboratories. Valuable solutions in many domains can probably result from the effort to build low-cost, high-rate, high-accuracy particle detectors required by the next generation of accelerators.”

This interview is adapted from the 2004 book “Infinitely CERN”, published to celebrate CERN’s 50th anniversary. Georges Charpak passed away in 2010 at the age of 86, read a tribute to him in the [CERN Courier](#).

Probing matter–antimatter asymmetry with AI

Using a cutting-edge AI algorithm, the CMS collaboration has obtained the first evidence of CP violation in the decay of the strange beauty meson into a pair of muons and a pair of electrically charged kaons

When we look at ourselves in a mirror, we see a virtual twin, identical in every detail except with left and right inverted. In particle physics, a transformation in which charge–parity (CP) symmetry is respected swaps a particle with the mirror image of its antimatter particle, which has opposite properties such as electric charge.

The physical laws that govern nature don't respect CP symmetry, however. If they did, the Universe would contain equal amounts of matter and antimatter, as it is believed to have done just after the Big Bang. To explain the large imbalance between matter and antimatter seen in the present-day Universe, CP symmetry has to be violated to a great extent. The Standard Model of particle physics can account for some CP violation, but it is not sufficient to explain the present-day matter–antimatter imbalance, prompting researchers to explore CP violation in all its known and unknown manifestations.

One way CP violation can manifest itself is in the “mixing” of electrically neutral mesons such as the strange beauty meson, which is composed of a strange quark and a bottom antiquark. These mesons can travel macroscopic distances in the Large Hadron Collider (LHC) detectors before decaying into lighter particles, and during this journey they can turn into their corresponding antimesons and back.

This phenomenon, called meson mixing, could be different for a meson turning into an antimeson versus an antimeson turning into a meson, generating CP violation. To see if that's the case, researchers need to count how many mesons or antimesons survive a certain duration before decaying, and then repeat the measurement for a given range of durations. To do so, they have to separate mesons from antimesons, a task called flavour tagging. This task is crucial to pinning down CP violation in meson mixing and in the interference between meson mixing and decay.

At a seminar held recently at CERN, the CMS collaboration at the LHC reported the first

evidence of CP violation in the decay of the strange beauty meson into a pair of muons and a pair of electrically charged kaons.

By deploying a new flavour-tagging algorithm on a sample of about 500 000 decays of the strange beauty meson into a pair of muons and a pair of charged kaons, collected during Run 2 of the LHC, the CMS collaboration measured with improved precision the parameter that determines CP violation in the interference between this meson's mixing and decay. If this parameter is zero, CP symmetry is respected. The new flavour-tagging algorithm is based on a cutting-edge artificial intelligence (AI) technique called a graph neural network, which performs accurate flavour tagging by gathering information from the particles surrounding the strange beauty meson and those being produced alongside it.

The collaboration then combined the result with its previous measurement of the parameter based on data from Run 1 of the LHC. The combined result is different from zero and is consistent with the Standard Model prediction and with previous measurements from CMS and the ATLAS and LHCb experiments.

Notably, the combined result is comparable in precision to the world's most precise measurement of the parameter, obtained by LHCb, a detector specifically designed to perform measurements of this kind. Moreover, the result has a statistical significance that crosses the conventional “3 sigma” threshold, providing the first evidence of CP violation in the decay of the strange beauty meson into a pair of muons and a pair of charged kaons.

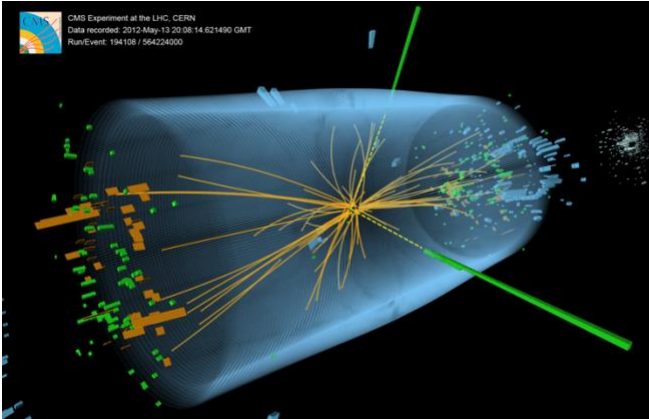
The result marks a milestone in CMS's studies of CP violation. Thanks to AI, CMS has pushed the boundary of what its detector can achieve in the exploration of this fundamental matter–antimatter asymmetry.

Find out more on the [CMS website](#).

CMS collaboration

CMS releases Higgs boson discovery data to the public

The collaboration has also made publicly available the software that it developed to search for the unique particle



CMS event display of a candidate Higgs boson decaying into two photons, one of the two decay channels that were key to the discovery of the particle. (Image: CERN)

As part of its continued commitment to making its science fully open, the CMS collaboration has just publicly released, in electronic format, the combination of CMS measurements that contributed to establishing the discovery of the Higgs boson in 2012. This release coincides with the publication of the Combine software – the statistical analysis tool that CMS developed during the first run of the Large Hadron Collider (LHC) to search for the unique particle, which has since been adopted throughout the collaboration.

Physics measurements based on data from the LHC are usually reported as a central value and its corresponding uncertainty. For instance, soon after observing the Higgs boson in LHC proton–proton collision data, CMS measured its mass as 125.3 plus or minus 0.6 GeV (the proton mass being about 1 GeV). But this figure is just a brief summary of the measurement outcome, a bit like the title of a book.

In a measurement, the full information extracted from the data is encoded in a mathematical function, known as the likelihood function, that includes the measured value of a quantity as well as its dependence on external factors. In the case of a CMS measurement, these factors encompass the calibration of the CMS detector, the accuracy

of the CMS detector simulation used to facilitate the measurement and other systematic effects.

A likelihood function of a measurement based on LHC data can be complex, as many aspects need to be pinned down to fully understand the messy collisions that take place at the LHC. For example, the likelihood function of the combination of CMS Higgs boson discovery measurements, which CMS just released in electronic format, has nearly 700 parameters for a fixed value of the Higgs boson mass. Among these, only one – the number of Higgs bosons found in the data – is the physics parameter of interest, while the rest model systematic uncertainties.

Each of these parameters corresponds to a dimension of a multi-dimensional abstract space, in which the likelihood function can be drawn. It is hard for humans to visualise a space with more than a few dimensions, let alone one with many. The new release of the likelihood function of the CMS Higgs boson discovery measurements – the first likelihood function to be made publicly available by the collaboration – allows researchers to get around this problem. With a publicly accessible likelihood function, physicists outside the CMS collaboration can now precisely factor in the CMS Higgs boson discovery measurements in their studies.

The release of this likelihood function, as well as that of the Combine software, which is used to model the likelihood and fit the data, marks a new milestone in CMS's decade-long commitment to fully open science. It joins hundreds of open-access publications, the release of almost five petabytes of CMS data on the CERN open-data portal and the publication of its entire software framework on GitHub.

Find out more on the [CMS website](https://cms.cern).

CMS collaboration

MoEDAL zeroes in on magnetic monopoles

The latest searches conducted by the MoEDAL experiment at the Large Hadron Collider considerably shrink the theoretical arenas in which the hunt for magnetic monopoles can continue



The MoEDAL detector. (Image: CERN)

The late physicist Joseph Polchinski once said the existence of magnetic monopoles is “one of the safest bets that one can make about physics not yet seen”. In its quest for these particles, which have a magnetic charge and are predicted by several theories that extend the Standard Model, the MoEDAL collaboration at the Large Hadron Collider (LHC) has not yet proven Polchinski right, but its latest findings mark a significant stride forward. The results, reported in two papers posted on the arXiv preprint server, considerably narrow the search window for these hypothetical particles.

At the LHC, pairs of magnetic monopoles could be produced in interactions between protons or heavy ions. In collisions between protons, they could be formed from a single virtual photon (the Drell–Yan mechanism) or the fusion of two virtual photons (the photon-fusion mechanism). Pairs of magnetic monopoles could also be produced from the vacuum in the enormous magnetic fields created in near-miss heavy-ion collisions, through a process called the Schwinger mechanism.

Since it started taking data in 2012, MoEDAL has achieved several firsts, including conducting the first searches at the LHC for magnetic monopoles produced via the photon-fusion mechanism and through the Schwinger mechanism. In the first of its latest studies, the MoEDAL collaboration sought monopoles and high-electric-charge objects (HECOs) produced via the Drell–Yan and photon-fusion mechanisms. The search was based

on proton–proton collision data collected during Run 2 of the LHC, using the full MoEDAL detector for the first time.

The full detector comprises two main systems sensitive to magnetic monopoles, HECOs and other highly ionising hypothetical particles. The first can permanently register the tracks of magnetic monopoles and HECOs, with no background signals from Standard Model particles. These tracks are measured using optical scanning microscopes at INFN Bologna. The second system consists of roughly a tonne of trapping volumes designed to capture magnetic monopoles. These trapping volumes – which make MoEDAL the only collider experiment in the world that can definitively and directly identify the magnetic charge of magnetic monopoles – are scanned at ETH Zurich using a special type of magnetometer called a SQUID to look for any trapped monopoles they may contain.

In their latest scanning of the trapping volumes, the MoEDAL team found no magnetic monopoles or HECOs, but it set bounds on the mass and production rate of these particles for different values of particle spin, an intrinsic form of angular momentum. For magnetic monopoles, the mass bounds were set for magnetic charges from 1 to 10 times the fundamental unit of magnetic charge, the Dirac charge (gD), and the existence of monopoles with masses as high as about 3.9 trillion electronvolts (TeV) was excluded. For HECOs, the mass limits were established for electric charges from 5e to 350e, where e is the electron charge, and the existence of HECOs with masses ranging up to 3.4 TeV was ruled out.

“MoEDAL’s search reach for both monopoles and HECOs allows the collaboration to survey a huge swathe of the theoretical ‘discovery space’ for these hypothetical particles,” says MoEDAL spokesperson James Pinfold.

In its second latest study, the MoEDAL team concentrated on the search for monopoles produced via the Schwinger mechanism in heavy-ion collision data taken during Run 1 of the LHC. In a unique endeavour, it scanned a decommissioned

section of the CMS experiment beam pipe, instead of the MoEDAL detector's trapping volumes, in search of trapped monopoles. Once again, the team found no monopoles, but it set the strongest-to-date mass limits on Schwinger monopoles with a charge between $2g_D$ and $45g_D$, ruling out the existence of monopoles with masses of up to 80 GeV.

"The vital importance of the Schwinger mechanism is that the production of composite monopoles is not suppressed compared to that of elementary ones, as is the case with the Drell–Yan

and photon-fusion processes," explains Pinfeld. "Thus, if monopoles are composite particles, this and our previous Schwinger-monopole search may have been the first-ever chances to observe them."

The MoEDAL detector will soon be joined by the MoEDAL Apparatus for Penetrating Particles, MAPP for short, which will allow the experiment to cast an even broader net in the search for new particles.

Ana Lopes

MADMAX at the forefront of the search for axions

The MADMAX experiment at CERN's North Area probes dark matter candidates with two new prototypes



MADMAX's new detector and the Morpurgo magnet in CERN's North Area. (Image: M.Cavazza/CERN)

Imagine a set of classic Matryoshka dolls, each containing a smaller doll inside it. MADMAX – not the movie, but an experiment hosted in CERN's largest experimental area, the SPS North Area – employs a similar technique to search for axions, a leading candidate for dark matter.

MADMAX (Magnetized Disk and Mirror Axion experiment) consists of multiple dielectric disks and a focusing mirror called the booster. Like the smaller dolls nestled within the larger ones, the booster is surrounded by an inner and outer vessel. There is a vacuum between these vessels to allow physics data taking at very low temperatures. Finally, the giant Morpurgo magnet serves as the outermost doll, providing the magnetic field essential for the experiment's operation. The Morpurgo magnet is the largest warm bore dipole in the world, with a 1.6-tesla

magnetic field, and is mainly used to test subdetectors of the ATLAS experiment.

In February and March this year, the two new prototypes of MADMAX have come into action to collect physics data at room temperature and, for the first time, at close to liquid helium temperature.

"We used Closed Booster 200, our new 200-mm-disk prototype, to collect physics data at room temperature, and Closed Booster 100, our 100-mm-disk prototype, to collect data close to the temperature of liquid helium, at around 10 K (about $-263\text{ }^{\circ}\text{C}$). At this temperature, the background thermal noise is lower than at room temperature, significantly increasing the sensitivity to axions," explains Pascal Pralavorio, a physicist from the MADMAX collaboration.

The collected data was discussed in the MADMAX collaboration meeting held at CERN in March this year. The data is currently being analysed and the collaboration looks forward to sharing its physics results in the future.

In the last decade, physicists have explored several experimental approaches, such as the CERN Axion Solar Telescope, to search for axions. To date, no experiment has succeeded in finding them. If axions are discovered, it would have profound implications for our understanding of both particle physics and cosmology. Firstly, it would validate the existence of a new particle predicted by theorists more than 40 years ago, confirming our understanding of fundamental forces in the

Universe. Secondly, since axions are considered a leading candidate for dark matter, their discovery could provide a direct explanation for this elusive substance that makes up a significant portion of the Universe.

MADMAX is a relatively young collaboration that started in 2017. Since 2020, CERN has provided the Morpurgo magnet to the experiment during technical stops, when the SPS beam is shut down. MADMAX benefits from a strong participation from across CERN for cryogenics, magnets, electrical power convertors, and safety and

operations. It is one of the few experiments at CERN that are well suited for tests independent of beam time, as it does not require a particle beam from an accelerator.

The MADMAX collaboration is currently discussing with the SPS Committee the possibility of using the Morpurgo magnet for another programme of data taking during the Long Shutdown 3 of CERN's accelerator complex, which will take place from 2026 to 2028.

Chetna Krishna

Hunting for millicharged particles at the LHC

The FASER and FORMOSA collaborations team up to test a demonstrator experiment to detect particles with a tiny electrical charge

The LHC family of experiments continues to grow. Alongside the four main experiments, a new generation of smaller experiments is contributing to the search for particles predicted by theories beyond the Standard Model, our current theory of particle physics. Recently, the FORMOSA demonstrator, which hunts for millicharged particles, has been installed in the cavern containing the FASER detector, 480 meters downstream from the ATLAS interaction point. It will now collect its first data.

Some theories predict the existence of millicharged elementary particles that would have a charge much smaller than the electron charge. If they exist, they would give clues to a theory beyond the Standard Model and could be considered as candidates for dark matter.

The FORMOSA demonstrator aims to prove the feasibility of the full experiment, which is intended to be installed in a proposed underground hall located about 620 metres away from the ATLAS interaction point. This experimental area – the Forward Physics Facility – is under study within the Physics Beyond Colliders initiative and is expected to host several experiments that will search for long-lived particles predicted by theories beyond the Standard Model. These particles would be produced by collisions at the centre of the ATLAS detector and would interact feebly with Standard Model particles. If approved, the experiments, among them the the proposed FASERv 2 and FLArE

experiments, could start taking data when the High-Luminosity LHC is switched on in 2029.

The FORMOSA demonstrator comprises scintillators. When interacting with a charged particle, the scintillators emit photons that are subsequently converted into an electrical signal. While cosmic muons or those from ATLAS collisions may also strike the scintillators, millicharged particles typically deposit much less energy into each layer, distinguishing them from muons that traverse the detector. “Initial studies with so-called no-beam data and source tests look already promising. This marks an important step towards achieving the goal to run the demonstrator this year and a great demonstration of the collaborative spirit of the projects within the Forward Physics Facility,” says project leader Matthew Citron from University of California, Davis.

Millicharged particles have become a particular focus of research in recent years. The MilliQan detector, located 33 meters away from the CMS interaction point, as well as MoEDAL-MAPP close to LHCb, started data taking during LHC Run 3. In 2020, a study carried out with a smaller demonstrator, MilliQan had ruled out the existence of millicharged particles for a range of masses and charges. Thanks to a higher volume of detection and its location in the far forward region of the LHC collisions, the FORMOSA experiment hopes to extend this search.

Kristiane Bernhard-Novotny

CERN 70th anniversary exhibition at Geneva Airport

Journey to the world of particle physics before jumping on your next flight



The exhibition at Geneva Airport celebrates CERN's 70th anniversary and will run until autumn 2024. (Image: CERN)

To honour its 70 years of contributions to scientific knowledge, technological innovation and

international collaboration, CERN has put together a rich and diverse programme, at CERN and across its Member States, Associate Member States and beyond. This programme includes exhibitions, the first of which can now be visited at Geneva Airport as part of a collaboration between the two organisations. Inaugurated on 2 May, the exhibition's three components will occupy the wall leading to the security check before entering the departure lounge, the "Panorama" terrace and the international terminal until autumn 2024.

Find out more on the "[CERN and its neighbours](#)" [website](#).

Zoe Nikolaidou

Unveiling the science of tomorrow: FCC Study takes centre stage at La Roche-sur-Foron exhibition

The Future Circular Collider team discussed the project's status and aspirations with a large number of attendees

The [Future Circular Collider \(FCC\) study](#) took centre stage at the International Fair of Haute-Savoie/Mont Blanc in La Roche-sur-Foron from 27 April to 6 May. An information booth, overflowing with interactive exhibits, captivating presentations and branded goodies, showcased the proposed research infrastructure's scientific potential, alongside the applications of particle physics research in everyday contexts.

The FCC study envisages a next generation particle collider that could succeed the LHC at CERN, currently the most powerful collider in the world. The FCC aspires to offer the broadest possible exploration of the Universe's mysteries via high-precision and high-energy studies of the elementary constituents of matter and the forces governing their interactions.

The information booth at the International Fair of Haute-Savoie/Mont Blanc provided a unique opportunity for residents and visitors to articulate

their views on the project and engage in constructive dialogue with the FCC team. Acknowledging the importance of transparency and community involvement, the project team is committed to openly addressing the hurdles inherent in such a monumental scientific endeavour. CERN's participation in the International Fair of Haute-Savoie/Mont Blanc, enhanced by the valuable help of volunteers from the FCC team, resulted in meaningful discussions with more than 2000 members of the local community on topics ranging from the required technological advancements to sustainability measures.

Zoe Nikolaidou

More information about the FCC study:

<https://fcc.web.cern.ch>

<https://fcc-faisabilite.eu>

Computer Security: WhiteHat & Zebra trainings are back

Vulnerabilities and weaknesses lurk all over the digital place: unprotected file uploads, generous SQL queries, unfiltered input fields, disclosed passwords... They are the entry point for cross-site scripting, remote code execution or root privilege escalation, and the first step towards fostering the patient-zero-like compromise of a server, a service or the whole of CERN; the first step towards losing protected, restricted or confidential information; the first step that can result in mild to serious reputational damage for the Lab.

A plethora of means exists to protect against that. On the one hand, the Computer Security team is [scanning for vulnerabilities and weaknesses](#) in the hope of detecting them early and mitigating them fast. On the other, hand-in-hand with the Computer Security team, you, as an excellent software developer and experienced programmer, [have followed the right courses](#) to put in place a secure software programming and code development life-cycle, including sound system architecture and choice of components, [apply best practices for managing and building your software](#) in a secure fashion, and are aware of [\(and can avoid!\) potential supply chain traps](#).

The CERN WhiteHat Challenge

While external students continue hacking into CERN and finding “juicy” stuff, we are glad to announce that the [WhiteHatChallenge](#) is back at CERN after a two-year hiatus. Designed as two half-days of training on [ethics and integrity](#), focusing on penetration testing and vulnerability scanning, it should bring you up to speed on detecting suboptimal configurations and weaknesses in your web services and websites. While penetration testing is a marathon that requires lots of training and practice, this

WhiteHat training should at least get you up and walking. It will cover the concepts for breaking into and abusing web applications, the use of the appropriate tools, and a Capture the Flag (CTF) tournament as a homework exercise to sharpen your skills. Hopefully it will give you a taste for becoming – after lots more fun training – an experienced penetration tester, hacker and participant in worldwide CTF tournaments! So, join us! All details for [afternoon 1](#) and [afternoon 2](#) can be found on Indico – no registration necessary.

Zebra Alliance Incident Response Exercise

And if you want to experience the pressure when the going really gets tough, see how incident response is conducted in reality. The Zebra Alliance has been hacked yet again (after previous attacks in [2022](#) and [2023](#))! And once again, it’s up to you and your peers in the room to figure out what happened. How did the attacker get in? What was their technique and intrusion vector? What’s their name (so the police can apprehend them)? No prior knowledge of security, incident response or even IT is needed. All you need is a laptop, the curiosity to dig and the eagerness to learn and tackle the challenge. As seats are limited, however, we kindly request that you [register on Indico](#). Have fun at both of these events! We hope to see you soon!

The Computer Security team

Do you want to learn more about computer security incidents and issues at CERN? Follow our [Monthly Report](#). For further information, questions or help, check [our website](#) or contact us at Computer.Security@cern.ch.

Announcements

Upcoming events

Check out this curated list of upcoming events for the CERN community

15–19 May | Knowledge Sharing | CERN Science Gateway | CineGlobe | EN & FR

[CineGlobe 2024](#)

19 May 17:00 | Knowledge Sharing | CERN Science Gateway | CERN70 | EN

[CERN70 public event: CERN - An extraordinary human endeavour](#) (more [CERN70](#) events)

23 May 14:00 | Knowledge Sharing | Online | HR Learning and Development | EN

[L&D Micro-talk 10 – Artificial Intelligence \(AI\) & The Future of Work](#)

24 May 11:00 | Knowledge Sharing | Online | CERN Alumni | EN

[CERN Alumni Virtual Company Showroom with the European Spallation Source](#)

30 May 12:15 | At CERN | Meyrin site | CERN Running Club | EN & FR

[52nd CERN Relay Race](#)

30 May 16:30 | Knowledge Sharing | Main Auditorium | CERN Colloquium | EN

[Quantum Gravity and Predictions for our Universe](#)

Registration now open

2–15 Jun | Accelerators | Sint-Michielsgestel (NL) | CERN Accelerator School | EN

[Mechanical & Materials Engineering for Particle Accelerators and Detectors](#)

10–12 Jun | Knowledge Sharing | Online | Sustainable HEP | EN

[Sustainable HEP 2024 – 3rd Edition](#)

11–13 Jun | Engineering | CERN | FDF | EN
[1st FPGA \(Field Programmable Gate Arrays\) Developers' Forum meeting](#)

22 Sep–5 Oct | Accelerators | Santa Susanna (ES) | CERN Accelerator School | EN

[Introduction to Accelerator Physics](#)

25 Sep–8 Oct | Physics | Peebles (UK) | ESHEP | EN

[2024 European School of High-Energy Physics](#)

This is a curated list of events relevant to the CERN community. More events are available here: home.cern/events

Indico also shows ALL events happening [today](#), [this week](#) and in a [calendar view](#).

If you would like your event to appear on an upcoming Bulletin events list, please contact bulletin-editors@cern.ch.

Building 60 status and next steps

With the renovation phase approaching, find out the next major steps for CERN's Main Building

Work on CERN's Main Building, Building 60, began in [April 2023](#) with a remediation phase to remove asbestos and other sources of pollution, including paint containing lead and PCBs. This remediation phase is now nearing completion and the renovation phase is about to begin.

With the renovation phase approaching, here are the next steps for the coming months:

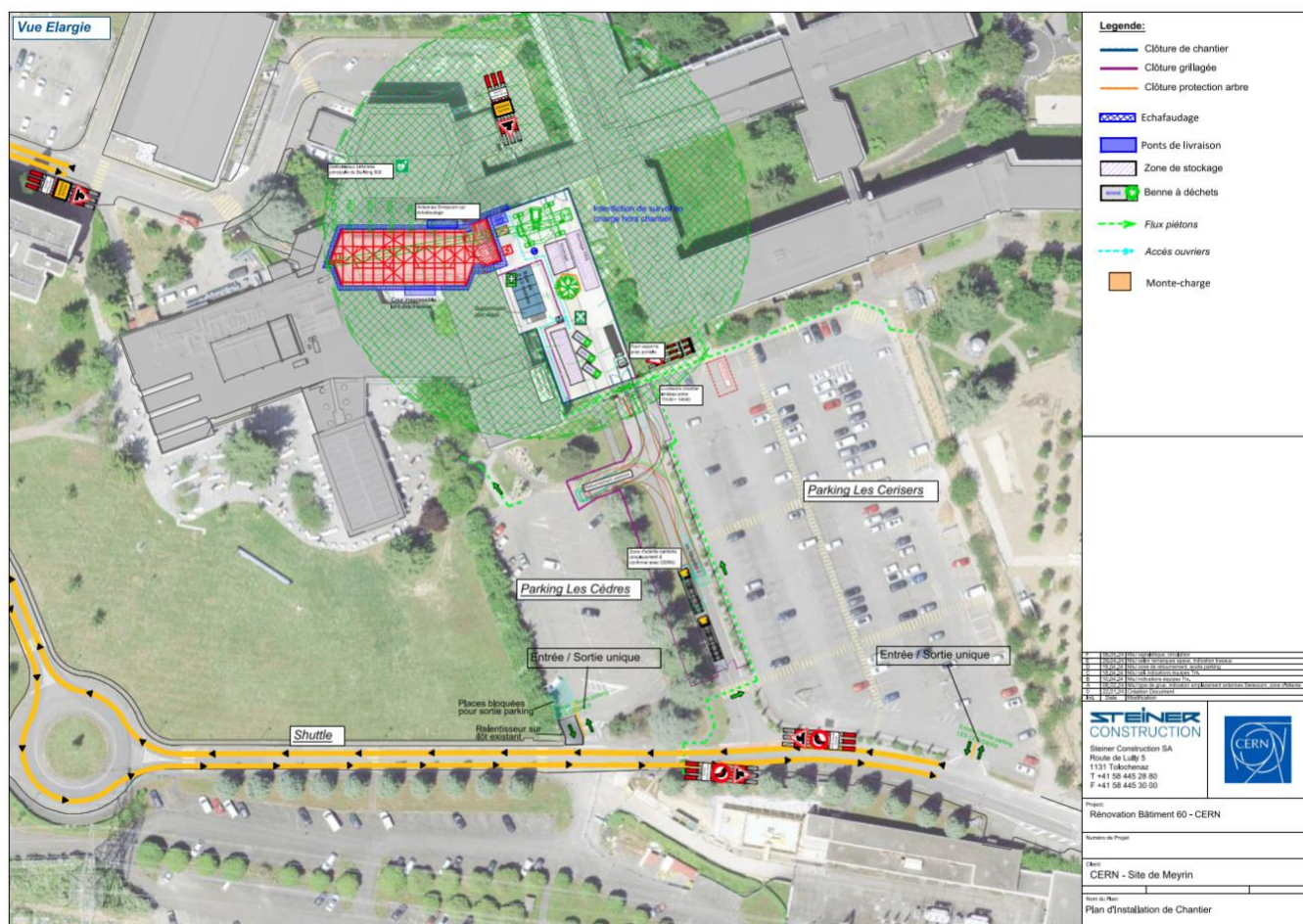
- **May and June: new construction site set up.** A new contractor team is replacing the heavy scaffolding with a lightweight model, reorganising site facilities into two levels and installing a crane. Access to the Les Cèdres and Les Cerisiers car parks will be modified, and Route Scherrer will be closed between the

construction site and Route Bohr (see plan below).

- **May and June: demolition of the 6th floor structure** (formerly the Charpak room). This phase will entail a higher risk of noise. Those working in the surrounding area will be kept informed of progress.
- **June and July: disruption of the mobile phone network around Building 60** while the mobile service antennae move to a new location, see

[OTG0149872](#) for details. Those working in the nearby area are encouraged to enable [VoLTE](#) and [Wi-Fi calls](#) on their devices.

- **July, August and September: beginning of construction work of Building 60**, primarily of the new 6th floor. This phase entails a moderate risk of noise.
- **October and November: installation of the facades.** This phase also entails a moderate risk of noise.



The SCE department would like to thank the CERN community, particularly those working in the area near Building 60, for their understanding during the period of disruption caused by this crucial and delicate project.

For more information, please see the [project webpage](#) or send an email to batiment.60@cern.ch

SCE department

Central heating turned off on 7 May

The central heating at the Meyrin and Prévessin sites is usually turned off around mid- to late May each year. Reducing the number of days of heating on campus was one of the measures included in

CERN's Energy Savings campaign launched in 2022.

In line with the Organization's long-term strategy for sustainability and its energy-saving policy, it has been decided to turn the heating on and off

according to the weather forecast instead of a defined regular calendar.

The weather forecast this year, particularly the overnight temperature, allows us to turn off the central heating on 7 May.

SCE department

Anticipating heatwaves

CERN's Organization-wide Heatwave Action Plan is on standby

The recent hot spell from 13–14 April was a reminder of the unpredictability of weather patterns and the challenge of anticipating them: a heatwave could occur at any time so CERN must always be ready to activate its Heatwave Action Plan, which was developed and first implemented in 2023. The CERN Heatwave Action Plan consists of a set of preventive and corrective measures designed to mitigate the effects of sustained high temperatures on members of the personnel during a heatwave. These measures, some of which depend on their compatibility with the needs of individual services, only apply during heatwave periods declared by official meteorological forecasts, which are monitored closely by the HSE unit. Heatwaves are declared by MeteoSwiss when average temperatures

exceeding 25 °C are forecast for at least three consecutive days.

The activation and end of the Heatwave Action Plan will be announced on the home.cern and hse.cern websites, as well as on posters displayed on restaurant screens (shown below) and in messages on the site entrance screens. All details of the Heatwave Action Plan can be found on the dedicated webpage: <https://hse.cern/hwap>.

The following [Beat the heat](#) tips provide advice on how to cope with high temperatures in the work environment, even outside of heatwave conditions, and can help avoid unpleasant symptoms.

Safety comes first, whatever the weather.

HSE unit

The VIP kitchen of Restaurant No. 1 gets a makeover

The GlassBox service will be moved to Caravelle room, on the first floor of Restaurant No. 2, for the duration of the renovation work

After the completion of the first stage of renovation work in Restaurant 1 in March 2023, a second phase of work will be carried out to completely refurbish the VIP kitchen.

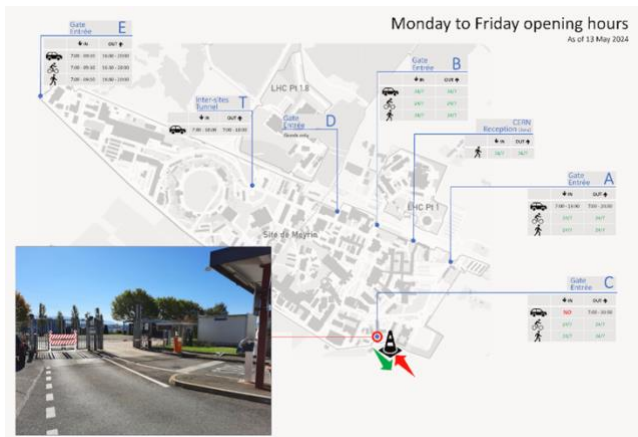
From 24 June to 8 September 2024, the GlassBox will be closed while its kitchen gets a makeover. The flooring, walls, ceiling, electrics and drains will all be renewed in order to bring them into line with new standards and optimise the space available, thereby improving the tools and working conditions of the teams there.

The equipment and light fittings will also be replaced with versions that are more energy efficient and better suited to the premises.

The GlassBox service will be moved to the Caravelle room, on the first floor of Restaurant No. 2, for the duration of the renovation work. Novae, our catering service provider, is doing its utmost to ensure that this move has as little impact as possible on its users.

Reservations can still be made via the usual channels: https://cern.service-now.com/service-portal?id=kb_article&n=KB0006220

From 13 May: Entrance C temporarily closed for incoming vehicles



Following damages caused by a car accident on 10 April, the opening hours of Entrance C (Route

Maxwell, on the Meyrin site) had been adapted in the past weeks, and the presence of a security guard was required.

Unfortunately, the repair requires parts that will take 13 weeks to arrive. As a result, Entrance C will be inaccessible to incoming vehicles as of Monday, 13 May 2024.

Entrance C remains open for outgoing vehicles during the usual hours from 7 a.m. to 8 p.m., and for pedestrians and bicycles 24/7.

The SCE and EN departments thank you for your understanding.

SCE department

The May/June issue of the CERN Courier is out

Uncovering the fundamental laws and constituents of the universe is not just a source of fascination for particle physicists. Engineers, chemists and materials scientists draw similar motivation in meeting the challenging and strange requirements of cutting-edge accelerator and detector technologies. The interdisciplinary materials, metrology and non-destructive testing section at CERN supports projects such as the HL-LHC magnet development with state-of-the-art equipment and analysis, and its services are increasingly in demand from projects outside – including the ITER fusion experiment (p37).

Also explored in depth in this issue are the next steps for the AWAKE plasma-wakefield

experiment (p25), the new-physics implications of neutrino masses (p29) and a lesser-known approach to quantum gravity called asymptotic safety (p43). Advanced triggers for the HL-LHC experiments (p8), the selection of the SHiP experiment for CERN's North Area (p7) and DESI's first cosmology results (p11) are among other highlights, along with the latest LHC results (p15), conference reports (p19), news in brief (p13), opinion (p49), reviews (p52), careers (p55) and more.

Read the digital edition of this new issue on CDS:
<https://cds.cern.ch/record/2896932>.